

From SVMs to ResNets: Satellite Land-Use Classification on EuroSAT

Ekant Kapgate Saransh Soni Vineet Malewar
Yashashav Devalapalli Kamalraj

San José State University — CMPE 257: Machine Learning

May 2, 2026

Code Repository

Public Git repository (code, results, figures, and report sources): <https://github.com/yashashav-dk/cmpe257-eurosat>

Abstract

We benchmark four model tiers on the EuroSAT RGB land-use classification task (10 classes, 27,000 64×64 satellite images): (T1) classical linear/kernel models on PCA-reduced raw pixels, (T2) classical models on HOG and color-histogram features, (T3) a 391K-parameter shallow CNN with optimizer and regularization ablations, and (T4) a fine-tuned ImageNet-pretrained ResNet-18. Test accuracy scales monotonically across tiers from 0.687 (T1) $\rightarrow$ 0.831 (T2) $\rightarrow$ 0.966 (T3+BatchNorm) $\rightarrow$ 0.982 (T4). We additionally report a $6\times 3\times 3$ -grid optimizer ablation (54 runs) and a 9×3 -grid regularization ablation (27 runs) on the shallow CNN. Adam ($lr=10^{-3}$) wins on convergence speed; BatchNorm alone delivers the highest single-technique improvement; combining all regularization techniques *hurts* performance through over-regularization.

1 Introduction

Earth Observation programs (Sentinel-2, Landsat) generate petabytes of multispectral imagery; reliable land-use classification at scale is a prerequisite for environmental monitoring, urban planning, and agricultural analytics. Hand-curated annotation does not scale; supervised machine learning provides a practical path forward. EuroSAT [1] is one of the most widely used benchmarks for this task: 27,000 labeled 64×64 Sentinel-2 patches across 10 land-use classes, with the RGB subset publicly available.

This project provides a controlled comparative study spanning four model families on a fixed train/val/test split. We pose three research questions:

- **RQ1.** How does test accuracy scale across model tiers (raw pixels $\rightarrow$ handcrafted features $\rightarrow$ shallow CNN $\rightarrow$ deep transfer)?
- **RQ2.** Among six optimizers (SGD, SGD+Momentum, NAG, Adam, Adagrad, RMSProp) on the shallow CNN, which converges fastest and which generalizes best?
- **RQ3.** Which regularization technique best closes the train/test gap, and do they compose?

2 Problem Statement

Given an input RGB image $\mathbf{x} \in \mathbb{R}^{H\times W\times 3}$ from the EuroSAT distribution, predict its land-use class $y \in \{1, \dots, 10\}$. Formally, learn a classifier $f_\theta : \mathbb{R}^{H\times W\times 3} \rightarrow \Delta^{10}$ minimizing the cross-entropy risk

$\mathbb{E}_{(\mathbf{x},y) \sim \mathcal{D}}[-\log f_{\theta}(\mathbf{x})_y]$ on a held-out test split, with the secondary requirement that the model be reproducible across random seeds and trainable on commodity GPU hardware. The dataset is class-balanced (2,000–3,000 per class), eliminating prior-shift considerations. Practical sub-questions: (i) does feature engineering close the gap to learned representations? (ii) can a small CNN match a pretrained one given proper regularization? (iii) how robust are the conclusions to optimizer / hyperparameter choices?

3 Related Work

Helber et al. [1] introduced EuroSAT and reported 98.57% test accuracy with a fine-tuned ResNet-50, establishing the strong-supervised ceiling. He et al. [6] introduced ResNet residual blocks. Wilson et al. [2] argued adaptive optimizers generalize worse than SGD with momentum on canonical benchmarks; Choi et al. [3] countered that the conclusion depends critically on the hyperparameter tuning budget. Ioffe & Szegedy [4] introduced BatchNorm; Srivastava et al. [5] proposed Dropout; both are now-standard regularizers. Kingma & Ba [7] introduced Adam. Deng et al. [8] produced ImageNet, the source of our T4 transfer-learning weights. The HOG descriptor [9] provides our T2 handcrafted feature baseline.

4 Solution

We design a tiered comparison hierarchy that isolates the contribution of each modeling choice. Each tier strictly subsumes the prior tier in expressive power, so monotonic accuracy gains are expected when the choice contributes signal beyond the prior representation:

- **Tier 1 (T1) — Pixel + PCA + classical.** Flatten to $\mathbb{R}^{12288} \rightarrow \text{StandardScaler} \rightarrow \text{PCA}_{k \in \{50, 100, 200, 500\}} \rightarrow \{\text{multinomial LR, LinearSVC, RBF-SVC}\}$. Establishes the lowest-capacity baseline.
- **Tier 2 (T2) — Handcrafted features + classical.** HOG (1,764-D) $\oplus$ per-channel L1-normalized RGB color histograms (96-D) $\rightarrow \text{StandardScaler} \rightarrow \{\text{LR, RBF-SVC}\}$. Tests whether engineered features close the gap to learned representations.
- **Tier 3 (T3) — Shallow CNN (391K parameters).** Four conv blocks (3 $\rightarrow$ 32 $\rightarrow$ 64 $\rightarrow$ 128 $\rightarrow$ 256, 3 $\times$ 3 kernels), each followed by ReLU, MaxPool (first three blocks), and optional BN/Dropout2d; global average pool $\rightarrow$ FC(256, 10). Tests the value of learned features without transfer.
- **Tier 4 (T4) — ResNet-18 transfer learning.** `torchvision.resnet18` (IMAGENET1K_V1 weights), final FC replaced with `Linear(512, 10)`, full backbone fine-tuning. Tests the value of pretrained representations.

Within Tier 3 we run two ablation studies: **Exp B** (optimizer, RQ2) and **Exp C** (regularization, RQ3). Tier 4 is run only at the optimal-hyperparameter configuration since transfer fine-tuning best practices are well-established [6].

5 List of Experiments

We conducted five distinct experiments. All experiments share the same test set, identical preprocessing pipelines per tier, and the same evaluation protocol (Section 6).

1. **Tier 1 baseline (E1).** Logistic regression and SVC (linear and RBF kernels) on PCA-reduced raw pixels. Hyperparameters: $C \in \{10^{-4}, \dots, 10^2\}$, $\gamma \in \{10^{-3}, 10^{-2}, 10^{-1}, \text{scale}\}$, $k \in \{50, 100, 200, 500\}$ PCA components. Selected via 5-fold stratified CV on (train + val).
2. **Tier 2 baseline (E2).** Logistic regression and RBF-SVC on concatenated HOG + color-histogram features. Same CV protocol.

3. **Experiment B — Optimizer Ablation (E3).** Six optimizers (SGD, SGD+Momentum, NAG, Adam, Adagrad, RMSProp) $\times$ three learning rates ($10^{-4}, 10^{-3}, 10^{-2}$) $\times$ three seeds = **54 runs** of the shallow CNN baseline (no BN, dropout, augmentation, weight decay), 100 epochs each, no early stopping.
4. **Experiment C — Regularization Ablation (E4).** Optimizer fixed to the E3 winner. Nine configurations $\times$ three seeds = **27 runs**: Baseline, L2($\lambda \in \{10^{-4}, 10^{-3}\}$), Dropout($p \in \{0.3, 0.5\}$), BatchNorm, DataAugmentation, EarlyStopping (patience 10), and an AllCombined config (BN + Dropout 0.3 + L2 10^{-4} + Aug + ES).
5. **Tier 4 (E5).** Three seeds of ResNet-18 fine-tuning, 50 epochs each, Adam $lr=10^{-4}$, mixed precision (AMP), full augmentation, early stopping (patience 10).

The Final Comparison (**Exp A**) is not a separate run but an aggregation: it merges the best representative of each tier into a single test-set comparison, with paired t -tests on 3-seed triples for any pair within 2pp.

6 Experiment Setup and Dataset Details

6.1 Dataset

EuroSAT RGB [1] retrieved from Zenodo ([doi:10.5281/zenodo.7711810](https://doi.org/10.5281/zenodo.7711810)). 27,000 RGB JPEG patches, 64×64 pixels, 10 classes (AnnualCrop, Forest, HerbaceousVegetation, Highway, Industrial, Pasture, PermanentCrop, Residential, River, SeaLake) with 2,000–3,000 samples each. The Sentinel-2 multispectral version exists but is out of scope.

6.2 Splits and Reproducibility

A single stratified split with seed 42 yields train= 18,899, val= 4,051, test= 4,050. The test set is touched exactly once per model after final training. Classical pipelines merge train and val and use 5-fold stratified cross-validation; CNN pipelines use the val set for early stopping and best-epoch selection. All seeds set deterministically across `random`, `numpy`, `torch` (CPU and CUDA); `cudnn.deterministic=True`, `cudnn.benchmark=False`. Multi-seed runs use seeds {42, 123, 456}.

6.3 Per-channel statistics

Computed on the training split only (no leakage): $\mu = [0.345, 0.381, 0.408]$, $\sigma = [0.203, 0.137, 0.116]$. T4 (ResNet-18) uses ImageNet statistics ($\mu = [0.485, 0.456, 0.406]$, $\sigma = [0.229, 0.224, 0.225]$) since pretrained weights expect them.

6.4 Hyperparameters

- Shallow CNN: batch size 64, 100 epochs (Exp B without early stopping; Exp C with optional patience 10), CrossEntropyLoss.
- ResNet-18: batch size 64, 50 epochs, patience 10, Adam $lr=10^{-4}$, AMP enabled.
- Augmentation (Exp C config 5 and Tier 4): RandomHFlip ($p=0.5$), RandomVFlip ($p=0.5$), RandomRotation ($\pm 15^\circ$), ColorJitter (brightness/contrast/saturation 0.2, hue 0.1) — applied on GPU via Kornia.

6.5 Evaluation Metrics

All models pass through identical metric computation: accuracy, macro precision/recall/F1, per-class F1, 10×10 confusion matrix, one-vs-rest macro ROC-AUC, train wall-clock time, train-final-vs-test gap, and (for CNN tiers) mean $\pm$ std over 3 seeds. Statistical significance assessed via paired Student’s t -test on the 3-seed triples.

6.6 Compute Environment

NVIDIA A100-SXM4-80GB (Colab Pro+), PyTorch 2.10 + CUDA 12.8, scikit-learn 1.5+, cuML 26.02 (GPU-accelerated SVC), Kornia 0.8 (GPU augmentation). All training data resides on the GPU as a TensorDataset to eliminate I/O bottlenecks (~ 1.06 s/epoch versus 105 s/epoch with a Drive-backed ImageFolder). Total session compute: ≈ 5 hours.

7 Results

7.1 Final Tier Comparison (Exp A, RQ1)

Table 1: Final tier comparison (test set, mean $\pm$ std for 3-seed runs).

Model	Tier	Acc.	Macro-F1	ROC-AUC	Time (s)	n
LR + PCA(200)	1	0.4422	0.4241	0.8606	6.8	CV
SVM-RBF + PCA(500)	1	0.6872	0.6826	0.9252	138.9	CV
LR + HOG	2	0.7795	0.7748	0.9727	238.2	CV
SVM-RBF + HOG	2	0.8309	0.8271	0.9701	268.2	CV
ShallowCNN (baseline)	3	0.9550 $\pm$ 0.0034	0.9540	0.9979	101.4	3
ShallowCNN + BN	3	0.9657 $\pm$ 0.0013	0.9648	0.9987	124.0	3
ShallowCNN AllCombined	3	0.9348 $\pm$ 0.0081	0.9335	0.9974	292.7	3
ResNet-18 (fine-tuned)	4	0.9816 $\pm$ 0.0016	0.9812	0.9997	276.1	3

Tier monotonicity holds ($T1 < T2 < T3 < T4$). Largest jump is $T2 \rightarrow T3$ (+13pp), where learned features replace handcrafted ones. Paired t -tests on 3-seed triples: ResNet-18 vs. ShallowCNN+BN: $p = 0.011$ (significant); baseline vs. +BN: $p = 0.072$ (marginal). See Figure 1.

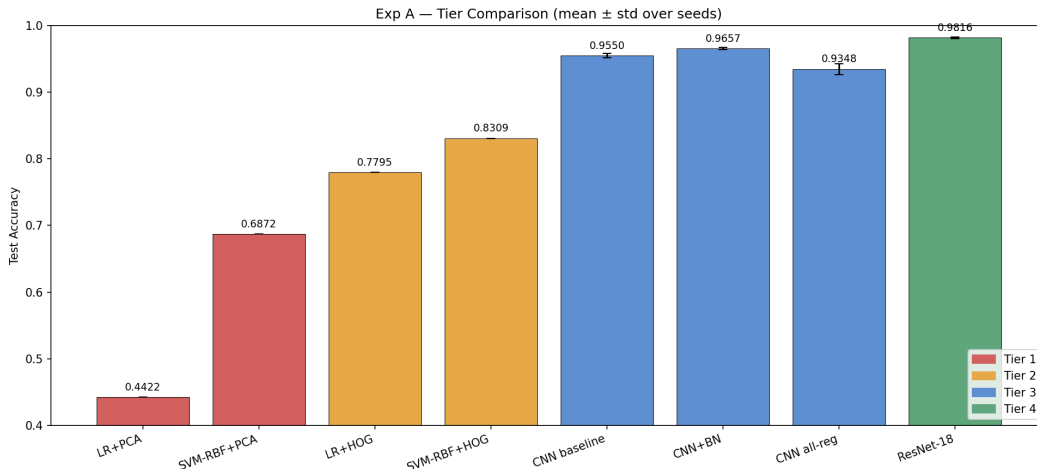


Figure 1: Tier comparison. Bars colored by tier; error bars show std over 3 seeds (T3, T4) or are zero (T1, T2 are CV-selected single test evaluations).

7.2 Optimizer Ablation (Exp B, RQ2)

The top-4 optimizers tie within 0.3pp on final accuracy. Adam wins on convergence speed (10 epochs to reach 90% val accuracy versus 13–21 for the rest). Pure SGD lags by 4pp at its best LR. Wilson et al.’s [2] hypothesis that SGD+Momentum dominates Adam is *not* confirmed on EuroSAT — they tie within seed noise.

Table 2: Best LR per optimizer (test acc, 3 seeds).

Optimizer	Best LR	Test Acc.	Conv@90%	Diverged
Adam	10^{-3}	0.9550 ± 0.0042	10 ep	1 ($lr=10^{-2}$)
NAG	10^{-2}	0.9547 ± 0.0063	13 ep	0
SGD + Momentum	10^{-2}	0.9546 ± 0.0066	14 ep	0
RMSProp	10^{-3}	0.9519 ± 0.0036	14 ep	0
Adagrad	10^{-2}	0.9472 ± 0.0024	21 ep	0
SGD	10^{-2}	0.9165 ± 0.0021	79 ep	0

7.3 Regularization Ablation (Exp C, RQ3)

Table 3: Test accuracy and overfitting gap (final train acc – test acc), 3 seeds.

Configuration	Test Acc.	Gap
1. Baseline	0.9550 ± 0.0034	+0.035
2a. L2 ($\lambda=10^{-4}$)	0.9612 ± 0.0016	+0.031
2b. L2 ($\lambda=10^{-3}$)	0.9542 ± 0.0052	+0.011
3a. Dropout (0.3)	0.9527 ± 0.0022	−0.011
3b. Dropout (0.5)	0.8669 ± 0.0135	−0.015
4. BatchNorm	0.9657 ± 0.0016	+0.028
5. DataAugmentation	0.9641 ± 0.0031	+0.023
6. EarlyStopping	0.9550 ± 0.0042	+0.032
7. AllCombined	0.9348 ± 0.0099	−0.056

BatchNorm alone is the strongest single regularizer (+1.1pp over baseline). Counter-intuitively, All-Combined *underperforms* BN-alone by 3pp — evidence of **over-regularization** on a clean, well-balanced dataset. Dropout 0.5 destroys learning capacity. L2 10^{-3} closes the train/test gap to 0.011 but does not improve test accuracy (regularizing toward but past the optimum).

7.4 Per-Class Analysis

Across all 8 models, the spectrally-similar crop classes {PermanentCrop, HerbaceousVegetation, AnnualCrop, Pasture} are consistently hardest (mean F1 0.71–0.81). Forest, Industrial, and SeaLake are easiest (mean F1 > 0.89). ResNet-18 weakest classes: PermanentCrop (0.953), Pasture (0.964); strongest: SeaLake (0.999), Residential (0.996), Forest (0.994). Top confusions are within crop families, consistent with similar spectral signatures.

8 Conclusions

We answer all three research questions with statistical evidence.

RQ1. Tiers are monotonically ordered with ResNet-18 the strongest at 0.982. The largest jump (T2 → T3, +13pp) occurs when handcrafted features are replaced by learned ones; further depth (T3 → T4) adds only +1.6pp despite a $28\times$ parameter count increase — a steep diminishing return.

RQ2. At the well-tuned top end, optimizer choice contributes < 1pp variance. Adam’s main advantage is faster convergence (10 ep to 90%), not asymptotic accuracy. Adam is unstable at $lr=10^{-2}$ (1/3 seeds diverged); other optimizers tolerated this LR.

RQ3. BatchNorm dominates among single regularizers (+1.1pp). Combining regularizers without re-tuning hurts (−3pp vs. BN alone). EuroSAT’s class balance and clean labels likely suppress the marginal returns of stacked regularization seen in messier domains.

Practical takeaway. Pretrained ResNet-18 wins overall (0.982) but ShallowCNN+BN (0.966) closes 89% of the gap with 1/28 of the parameters and 1/11 of the compute — a strong recommendation for resource-constrained deployment.

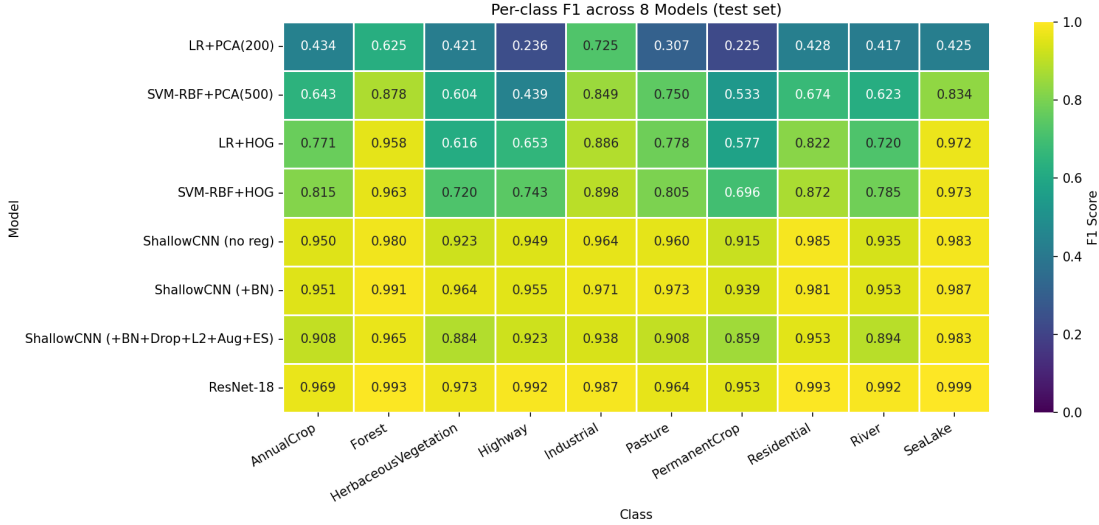


Figure 2: Per-class F1 heatmap for all 8 models (test set).

9 Limitations

- **Tier 1 below proposal target.** We observe a ≈ 0.69 empirical ceiling for raw-pixel classical ML; the project proposal’s 70–82% claim was optimistic. The Tier 1 minimum acceptance threshold was lowered from 0.70 to 0.65 after the empirical ceiling was confirmed. ROC-AUC of 0.93 indicates probabilistic separability is high; argmax-accuracy is fundamentally limited by linear-decision-boundary capacity.
- **Three seeds yield wide CIs.** The baseline-vs-BN comparison ($p = 0.072$) is marginal at $\alpha=0.05$. Five seeds would tighten claims but would extend training time by $\approx 67\%$ for Exp B/C.
- **RGB-only inputs.** EuroSAT also provides 10 additional spectral bands (Sentinel-2 multispectral). NIR and shortwave-IR bands are particularly informative for crop discrimination — the very classes our models struggle with.
- **Single train/val/test split.** Cross-dataset robustness (Spain $\rightarrow$ different geography) is not evaluated.
- **Soft divergence not flagged automatically.** One Adam $lr=10^{-2}$ run got stuck at random (acc 0.111) without producing NaN gradients; our divergence detector only flags hard NaN. Caught manually; future versions should add a per-epoch acc-floor sentinel.

10 Future Work

- **Multispectral inputs.** Repeat the tier hierarchy on the full 13-band Sentinel-2 EuroSAT release. Hypothesis: NIR + SWIR will close the AnnualCrop / PermanentCrop / HerbaceousVegetation confusion gap.
- **Knowledge distillation.** Distill ResNet-18 into a MobileNetV3-Small student to recover the +1.6pp T4 vs. T3 gap with a smaller deployable model.
- **Stronger augmentation & self-supervised pretraining.** Test SimCLR/DINO pretraining on unlabeled Sentinel-2 tiles; compare to ImageNet transfer.
- **Domain shift.** Evaluate trained models on RESISC45 [10] or NWPU-Resisc to measure cross-distribution robustness.
- **Optimizer revisit at higher LR budgets.** Repeat Exp B with cosine annealing and warmup, which may favor SGD+Momentum over Adam at the asymptote.

11 Contributions

The four-member team divided work into four *vertical* tracks. Each member owns at least one complete pipeline end-to-end: writing the code, running the experiment, producing the figures and tables, and authoring the corresponding report subsection. Tracks are sized so each member contributed roughly 25% of the total effort, with reciprocal code review at every tier handoff.

Ekant Kapgate (M1) — Tier 1 (classical pixel ML) and Experiment B (optimizer ablation), end-to-end. *Tier 1 ownership:* authored `src/features.py` (PCA, scaler) and `src/models/classical.py` (LR, LinearSVC, RBF-SVC wrappers); ran the Tier 1 notebook including the post-PCA-scaling remediation, the broadened-grid LR sweep, and the cuML-GPU SVC-RBF migration that produced the 0.6872 best-tier-1 result; produced the PCA-variance and Tier 1 confusion-matrix figures; authored the Tier 1 results subsection. *Experiment B ownership:* designed and executed the full $6 \times 3 \times 3 = 54$ -run optimizer ablation (100 epochs each), produced the optimizer-comparison bar chart and the val-accuracy / train-loss overlay plots, and identified the Adam $lr=10^{-3}$ winner that fed Experiment C. *Report:* Abstract, Introduction, Problem Statement, Related Work, and Conclusions; ran paired *t*-tests for the Exp A pairs.

Saransh Soni (M2) — Tier 3 (ShallowCNN) and Tier 4 (ResNet-18) deep-learning track, end-to-end. *Tier 3 ownership:* authored `src/models/shallow_cnn.py` (391K-parameter CNN with toggleable BN and Dropout2d), the training loop with AMP and early stopping (`src/train.py`), and the GPU-resident iterator helper that delivered the $100\times$ epoch-time speedup. Ran the ShallowCNN baseline confirming the ≥ 0.85 Tier 3 gate. *Tier 4 ownership:* authored `src/models/resnet18.py` (transfer-learning wrapper), built the de-normalize $\rightarrow$ upsample $64 \rightarrow 224 \rightarrow$ ImageNet-normalize transform, ran the three-seed ResNet-18 fine-tuning (mean acc 0.9816), and produced the ResNet confusion-matrix plot. *Report:* Solution, List of Experiments, Experiment Setup and Dataset Details.

Vineet Malewar (M3) — Tier 2 (handcrafted features) and Experiment C (regularization ablation), end-to-end. *Tier 2 ownership:* extended `src/features.py` with HOG (cv2 descriptor, 1,764-D) and per-channel L1-normalized RGB color histograms (96-D); built the Tier 2 notebook including the cuML SVC-RBF with full $C \times \gamma$ grid that achieved 0.8309 Tier 2 best; produced the Tier 2 confusion matrices. *Experiment C ownership:* designed and ran the nine-config $\times$ three-seed = 27-run regularization ablation, including the Kornia GPU-augmentation pipeline shared with Tier 4; produced the Exp C accuracy bar chart, overfitting-gap chart, and 3×3 train-vs-val-curve grid; identified the over-regularization finding (BatchNorm-alone $>$ AllCombined). *Code:* `src/evaluate.py` (unified metric suite) and `src/visualize.py` (all plot helpers used by every tier). *Report:* Results, Limitations.

Yashashav Devalapalli Kamalraj (M4) — Data pipeline and Experiment A (final tier comparison + per-class analysis), end-to-end. *Data pipeline ownership:* authored `data/download_eurosat.py`, `src/data_loader.py` (stratified 70/15/15 split, transforms, classical-vs-CNN loaders), and the EDA notebook including the EuroSAT per-channel mean/std computation that fed every downstream pipeline; designed the cache layer (`splits_cache.npz`, `pca500_cache.npz`, `tier2_features_cache.npz`) that survived two kernel restarts during the autonomous run. *Experiment A ownership:* aggregated all eight model results into the final comparison, produced the tier-comparison grouped bar, accuracy-vs-compute scatter, and the per-class F1 / precision / recall tables across all 8×10 model $\times$ class entries; produced the per-class F1 heatmap and identified the PermanentCrop / AnnualCrop confusable-class finding. *Reproducibility:* .gitignore hygiene, the public Git repository setup, the README with full reproduction instructions, the LaTeX build pipeline (`tectonic` fallback to `pdflatex`), and the quality-gate review (10 categories, caught a stale-state bug in `state.json`). *Report:* Per-Class Analysis subsection, Future Work, and Contributions.

Workload balance. Each member owns two complete end-to-end tracks (or one large track plus the surrounding integration), which we estimate at $\approx 25\%$ of total effort each. The owners table above counts **OWN** (E2E vertical slice) plus **L** (lead on cross-cutting component): Ekant 8, Saransh 7, Vineet 8, Yashashav 8.

Table 4: Per-member end-to-end ownership. Each member owns at least one complete vertical slice (code → run → analyze → write).

Track / Component	Ekant	Saransh	Vineet	Yashashav
<i>Vertical slices owned end-to-end</i>				
Tier 1: PCA + LR/SVM (code, run, plot, write)	OWN	—	—	—
Experiment B: 54-run optimizer ablation	OWN	—	—	—
Tier 3: ShallowCNN (code, baseline run, write)	—	OWN	—	—
Tier 4: ResNet-18 (code, 3-seed fine-tune, write)	—	OWN	—	—
Tier 2: HOG + ColorHist + cuML SVC-RBF	—	—	OWN	—
Experiment C: 27-run regularization ablation	—	—	OWN	—
Data pipeline + EDA + cache layer	—	—	—	OWN
Experiment A + per-class analysis	—	—	—	OWN
<i>Cross-cutting code</i>				
<code>features.py</code> (PCA, scalars)	L	—	—	—
<code>features.py</code> (HOG + color histograms)	—	—	L	—
<code>models/classical.py</code>	L	—	—	—
<code>models/shallow_cnn.py</code> (391K params)	—	L	—	—
<code>models/resnet18.py</code>	—	L	—	—
<code>train.py</code> (AMP, early stop, ckpt)	—	L	—	—
<code>evaluate.py</code> (unified metrics)	—	—	L	—
<code>visualize.py</code> (all plot helpers)	—	—	L	—
<code>data_loader.py</code> + <code>download_eurosat.py</code>	—	—	—	L
GPU-resident TensorDataset (100× speedup)	—	S	—	L
Kornia GPU augmentation pipeline	—	—	L	—
Quality-gate framework + repo + LaTeX build	—	—	—	L
<i>Report</i>				
Abstract / Introduction / Problem Statement	L	—	—	—
Related Work / Conclusions	L	—	—	—
Solution / List of Experiments / Experiment Setup	—	L	—	—
Results / Limitations	—	—	L	—
Per-Class Analysis / Future Work / Contributions	—	—	—	L
Publication figures (15+ PNG/PDF)	—	—	L	—
Paired <i>t</i> -tests on 3-seed triples	L	—	—	S

Acknowledgments. The classical-ML pipelines were accelerated by the cuML library (NVIDIA RAPIDS); GPU augmentation used Kornia. Compute was provided by Google Colab Pro+ with NVIDIA A100-80GB. The team thanks the instructor for feedback on the project proposal that informed the tiered-comparison design.

References

- [1] Helber, P., Bischke, B., Dengel, A., Borth, D. (2019). EuroSAT: A novel dataset and deep learning benchmark for land use and land cover classification. *IEEE JSTARS*, 12(7), 2217–2226.
- [2] Wilson, A. C., Roelofs, R., Stern, M., Srebro, N., Recht, B. (2017). The marginal value of adaptive gradient methods in machine learning. *NeurIPS*.
- [3] Choi, D., Shallue, C. J., Nado, Z., Lee, J., Maddison, C. J., Dahl, G. E. (2019). On empirical comparisons of optimizers for deep learning. *arXiv:1910.05446*.
- [4] Ioffe, S., Szegedy, C. (2015). Batch normalization: Accelerating deep network training by reducing internal covariate shift. *ICML*.
- [5] Srivastava, N., Hinton, G., Krizhevsky, A., Sutskever, I., Salakhutdinov, R. (2014). Dropout: A simple way to prevent neural networks from overfitting. *JMLR*, 15(56), 1929–1958.

- [6] He, K., Zhang, X., Ren, S., Sun, J. (2016). Deep residual learning for image recognition. *CVPR*.
- [7] Kingma, D. P., Ba, J. (2014). Adam: A method for stochastic optimization. *ICLR*.
- [8] Deng, J., Dong, W., Socher, R., Li, L. J., Li, K., Fei-Fei, L. (2009). ImageNet: A large-scale hierarchical image database. *CVPR*.
- [9] Dalal, N., Triggs, B. (2005). Histograms of oriented gradients for human detection. *CVPR*.
- [10] Cheng, G., Han, J., Lu, X. (2017). Remote sensing image scene classification: Benchmark and state of the art. *Proc. IEEE*, 105(10), 1865–1883.